

Quantum States Hidden in Linguistics

Xinmin Zhao, Feng Zhang*

*College of Foreign Languages, Terahertz Technology Innovation Research Institute,
Terahertz Spectrum and Imaging Technology Cooperative Innovation Center, Shanghai
Key Lab of Modern Optical System, University of Shanghai for Science and Technology,
Shanghai 200093, People's Republic of China*

Corresponding Email: fzhang@usst.edu.cn

Abstract: In recent years, the application of quantum mechanics in life sciences has given rise to quantum biology, which deeply investigates the impact of quantum effects on biological processes, such as quantum transfer in photosynthesis and quantum coherence phenomena in bird navigation. However, the connection between quantum mechanics and the social sciences, particularly linguistics, has not been widely explored. This paper draws analogies between the discrete nature, superposition states, and entanglement phenomena in quantum mechanics, and the discreteness, ambiguity, and context-dependence in linguistics, revealing interdisciplinary similarities between the two fields. This analogy provides a new perspective for understanding the uncertainty and complexity involved in human language processing, and suggests the possible existence of macroscopic quantum states. Future research may uncover the impact of quantum effects on higher cognitive functions and promote a deeper integration of quantum physics and social sciences.

Keywords: Quantum Mechanics, Linguistics, Discreteness, Superposition, Quantum Entanglement, Interdisciplinary Research

1. Introduction

Although significant progress has been made in the intersection of quantum mechanics and life sciences (Ball, 2011; Cai, 2016; Cao et al., 2020; Kim et al., 2021; Vattay et al., 2014), its application in the social sciences, particularly in linguistics, remains largely unexplored. Research in quantum biology has revealed the key role of quantum effects in biological processes, such as quantum transfer in photosynthesis (Trixler, 2013) and quantum coherence phenomena in bird navigation (Levy, 2021). Linguistics, as a discipline bridging biology and social sciences, relies on the brain's neural mechanisms (Bowern, 2015; Galofaro et al., 2016; Hruschka et al., 2015) while also encompassing social factors like culture and history. Its complexity and biological foundation make it one of the most natural-science-like social sciences, highlighting the significance of exploring its connection to quantum mechanics.

The complexity of linguistics is not only rooted in its biological basis but also in its deep ties to human cognition and social culture (Diamond, 2011; Fitch, 2007; Greenberg, 1959; Wuethrich, 2000). Language is the primary tool for humans to express thoughts and exchange information, and its structure and use are influenced by a variety of factors, including biology, psychology, and sociology. Similar to the life sciences, linguistics also involves substantial systemic and rule-based elements, but it simultaneously exhibits significant flexibility and uncertainty. For this reason, linguistic research is inherently interdisciplinary. The rise of interdisciplinary studies in recent years has led to the gradual application of quantum mechanics concepts to the understanding of linguistic phenomena. For example, quantum superposition has been used to explain ambiguity and semantic vagueness in language, while quantum entanglement provides a framework for understanding the close interrelations of words within context.

The similarities between quantum mechanics and linguistics manifest at multiple levels. Firstly, both fields involve the study of the smallest units within complex systems: in quantum mechanics, these units are quantum particles, whereas in

linguistics, they are phonemes, words, and grammatical rules. Secondly, concepts such as uncertainty and superposition in quantum mechanics (Schrodinger, 1944) offer new perspectives for understanding ambiguity and polysemy in language. This perspective allows us to comprehend, at a more abstract and fundamental level, how language reflects human cognitive characteristics. Moreover, the concept of quantum entanglement provides a new framework for explaining the complex relationship between words and context in language, allowing linguistic research to benefit from the theoretical tools of quantum physics in exploring how language constructs intricate semantic networks through the interrelation of its elements.

In addition, linguistics involves not only the internal structure of language (such as phonology, morphology, and syntax) but also the linguistic context and social background (such as pragmatics and sociolinguistics). This dual nature gives linguistic research a unique perspective, enabling the analysis of language phenomena from both individual cognitive and social interaction levels. The wave-particle duality in quantum mechanics (Koch, 2017) provides inspiration for understanding the dual nature of linguistics: language is both a product of internal cognition and a tool for social interaction. By drawing an analogy to the dual manifestations of particles and waves in quantum systems, we can better understand the multi-layered properties of language.

Therefore, applying the concepts of quantum mechanics to linguistic research offers new tools for understanding the complexity of language, particularly when addressing issues of ambiguity, polysemy, and context dependence. Core principles of quantum mechanics, such as superposition, measurement, and entanglement, serve as valuable analogies for similar phenomena in language. These analogies not only enhance our understanding of language but also provide new possibilities for the integration of linguistics with other disciplines. In the future, the intersection of quantum mechanics and linguistics may open new research directions, revealing the potential quantum characteristics of higher cognitive functions and further advancing the interdisciplinary fusion of natural and social sciences.

2. The Manifestation of Discreteness in Language and Other Systems

Any process of development or evolution from simple to complex can find its initial building blocks, all of which possess discreteness. Scientists created quantum to describe the smallest units—whether at the energy level or the material level. In other words, objects that can be described by quantum mechanics can be considered discontinuous—discrete—one of the three main features of quantum theory. The discreteness in quantum mechanics is manifested in the quantization of microscopic physical quantities, such as energy and angular momentum, which can only take specific discrete values, rather than varying continuously. This characteristic can also be analogized to the structural discreteness in language. The vocabulary and grammar systems are inherently discrete, for example, vocabulary consists of discrete phonemes or letters, and sentences are organized by discrete grammatical rules.

Discreteness can be observed in different languages. For example, in Chinese, the character is the smallest unit, and each character contains discrete subunits like strokes; in English, the letter is the smallest unit, and words are formed by the combination of letters. This combinatory approach is similarly reflected in languages like German and French. In Chinese, the meaning of words is typically determined by word order or collocations, and these combinations have clear boundaries, demonstrating discrete characteristics. English, on the other hand, forms vocabulary and sentences through affixes, word order, and other methods, with each component having a relatively independent and clearly defined meaning, ensuring the discrete structure of the language. This discreteness in language exists not only at the vocabulary level but also at the grammatical level. For example, English tense changes are reflected through changes in verb forms, with each change being discrete, having specific forms and rules; in Chinese, tense is expressed through time adverbs or contextual relationships, also showing clear grammatical boundaries. This discrete nature of grammatical systems gives the language structure a clear hierarchy, enabling the expression of infinite meanings through a finite set of grammatical rules. For instance, in Chinese, the

sentence "我昨天去了商店" ("I went to the store yesterday") contains discrete units: "我" (I), "昨天" (yesterday), "去了" (went), "商店" (store), each with independent meanings. Similarly, in English, the corresponding sentence "I went to the store yesterday" is composed of discrete components: "I", "went", "to", "the", "store", "yesterday", each having a clear meaning and not subject to continuous variation. In Arabic, in the sentence "ذهبت إلى المتجر أمس" ("I went to the store yesterday"), "ذهبت" (I went), "إلى" (to), "المتجر" (store), and "أمس" (yesterday) are all independent units, forming a complete sentence structure. In Japanese, the sentence "私は昨日店に行きました" ("I went to the store yesterday") expresses the same meaning, where "私" (I), "は" (topic marker), "昨日" (yesterday), "店" (store), "に" (direction particle), and "行きました" (went) are all discrete language units.

The origin of discreteness can be traced back to human cognition of the world. We tend to divide the world into independent, understandable parts, and this cognitive tendency is reflected in various fields. For example, in quantum mechanics, scientists use quantum to describe the smallest units of energy; in linguistics, humans divide and organize language through phonemes, vocabulary, and grammar. In biology, genes are considered the basic units of life, and genes form the various characteristics of organisms through discrete combinations; in computer science, bits are the smallest units of information, and through the combination of bits, complex data and programs are formed. In Arabic, the construction of language also reflects discreteness, with letters combined with vowel symbols to form complex vocabulary structures. Similarly, in Japanese, the combination of kana and kanji demonstrates a clear hierarchical structure in language composition. These examples all indicate that discreteness is a fundamental way for humans to understand and construct complex systems.

Discreteness allows for the possibility of transformation and evolution between languages. For example, Mongolian, Tibetan, and Uighur have different writing systems, but they can all be translated and transformed through discrete language units, enabling communication between different cultures. Chinese and Japanese also have an

evolutionary relationship, with Japanese kana and kanji combining partly inherited from Chinese writing systems. This evolution is a direct manifestation of linguistic discreteness. Similarly, languages such as English, French, German, and Spanish all derive from Latin, and through the evolution and branching of Latin, each language has formed its own unique linguistic system, with each language composed of discrete building units, enabling a certain degree of mutual conversion and learning between them. These examples demonstrate the important role of discreteness in linguistic evolution and mutual influence.

By comparing the discreteness in different languages, it can be observed that although there are differences in specific expression methods and construction rules, fundamentally, languages are all composed of discrete units, and these units are combined according to certain rules to form complex semantic structures. This discreteness allows language systems to possess a high degree of flexibility and expandability, enabling them to meet the ever-changing communication needs. The analogy between the discreteness in quantum theory and the hierarchical structure in linguistics reveals an interesting interdisciplinary phenomenon: whether in the physical world or the world of language, the construction of complex systems relies on the smallest, discrete building units, and these units evolve from simple to complex through different combinations. This universally present characteristic of discreteness helps us understand the formation and essence of various complex phenomena.

3. Quantum Superposition and Linguistic Uncertainty: An Interdisciplinary Analogy

Superposition is a profound concept in quantum mechanics, meaning that a quantum system can simultaneously exist in multiple mutually exclusive states, and only when a measurement is made does it "collapse" into a specific state. For example, in the famous Schrödinger's cat experiment, before the box is opened, the cat is both dead and alive, but when the observer opens the box to measure, the cat "collapses" into either a "dead" or "alive" state. This discovery challenged the linear causal view of classical

physics and showcased the unique coexistence of uncertainty and possibility in the microscopic world.

In linguistics, the concept of superposition can also be reflected in ambiguous sentences and fuzzy semantics. For instance, the English sentence “Visiting relatives can be annoying” can be interpreted in two ways: "Going to visit relatives is annoying" or "The visiting relatives are annoying," much like a quantum superposition, with the exact meaning needing to be determined by the context. In Chinese, the sentence "他打了电话给妈妈" ("He made a call to his mom") contains the word “打” (to strike or to dial), which could mean "dialing" or "hitting," and in the absence of clear context, this sentence also exhibits a superposition-like feature. In Arabic, the sentence "هو يكتب الكتاب" ("He is writing the book") can be interpreted as "He is writing a book" or "He is writing a letter," with the exact meaning depending on the context. Additionally, the use of vague language can also be seen as a superposition phenomenon. For example, in French, the word “beaucoup” (a lot) does not specify an exact quantity and only in a specific context can the scope of its meaning be determined. Similarly, in Japanese, the word “たくさん” (a lot) also relies on context to define its precise meaning. Vagueness, much like the superposition state in quantum systems, only collapses into a specific meaning through contextual measurement.

The concept of quantum superposition can also be used to explain uncertainty in grammar. In some languages, the structure and word order of a sentence can lead to multiple grammatical interpretations. For example, in English, the sentence “Flying planes can be dangerous” can be understood as “Flying planes are dangerous” or “Planes that are flying can be dangerous,” with this grammatical ambiguity resembling quantum superposition, and the exact interpretation needing to be determined by the context. In Chinese, the sentence "他把书放在桌子上" ("He placed the book on the table") also has a similar situation, where “桌子上” (on the table) can refer to the location of the action (placing) or the location of the object (book), and this grammatical uncertainty can only be clarified once the context is provided. Moreover, in Arabic, the

word order and morphological changes can lead to multiple syntactical possibilities. For instance, "هو يكتب" ("He writes") can be understood as "He is writing" or "He is recording," with the precise interpretation relying on additional contextual information.

From the perspective of different languages, the phenomenon of superposition is evident in polysemy, ambiguous sentences, and fuzzy language expression. Whether in English, Chinese, French, Japanese, or Arabic, there are examples. For instance, in English, the word "bark" can refer to "tree bark" or "a dog's bark," in French, "livre" can mean "book" or "one pound," in Chinese, "行" can mean "okay," "to walk," or "behavior," in Japanese, "さくら" (sakura) can refer to the cherry blossom in nature or a person named Sakura, and in Arabic, "سلام" (salaam) can mean both "peace" and be used as a greeting ("hello"). In German, "Schloss" can refer to "castle" or "lock," and the exact meaning, like a quantum system, must be determined through context. These words, when isolated from context, are like quantum systems in superposition, where their meaning only "collapses" into a specific interpretation when the context is clarified.

The phenomenon of polysemy and ambiguity in different languages reflects their differences and common patterns in semantic structure. For example, English and German, both belonging to the Indo-European language family, typically exhibit polysemy involving different interpretations between physical or abstract concepts, such as "bark" and "Schloss." Chinese, belonging to the Sino-Tibetan language family, has many words like "行" that possess broad semantic flexibility, and this polysemy often depends on specific sentence structures and context for differentiation. Arabic and French also have unique features in their polysemy phenomena. For instance, "سلام" (salaam) in Arabic carries an abstract meaning of peace in a religious or cultural context and also functions as a greeting in everyday language, while "voler" in French shows a shift between a concrete action (to steal) and an abstract behavior (to fly). By comparing the polysemy, ambiguity, and grammatical uncertainty phenomena across different languages, we can summarize a few patterns: First, polysemy in language is often

related to cultural background and historical development. Different language systems, due to their distinct cultures and histories, form their own unique ways of expressing polysemy. Second, context plays the role of "measurement" in all languages, determining the superposition state of language into a specific meaning. Finally, the uncertainty and flexibility of language provide the ability to express complex thoughts and emotions, which, like the concept of state superposition in quantum mechanics, demonstrates the diversity and depth of human thought.

Table 1. Comparison of Linguistic Uncertainty Across Different Language Families.

Language Family	Polysemy Example and Analysis	Ambiguity in Sentences and Analysis	Ambiguity in Semantics and Analysis
Chinese Family	打 (dǎ) can mean "to strike" (e.g., 打架 "fight"), "to make" (e.g., 打家具 "make furniture"), or "to knit" (e.g., 打毛衣 "knit a sweater"). Without context, the meaning of 打 is in a superposition of possibilities and only collapses when combined with specific context like "他在打 (击打) 鼓" (He is drumming) or "她在打 (编织) 围巾" (She is knitting a scarf).	“鸡不吃了” (Jī bù chī le) can be interpreted as "The chicken doesn't eat (food) anymore" or "I don't eat chicken anymore." Without further context, the sentence remains ambiguous, similar to a quantum superposition, and only becomes definite with additional context such as “鸡吃饱了, 鸡不吃了” (The chicken is full, it doesn't eat anymore) or “我减肥, 鸡不吃了” (I'm dieting, I don't eat chicken anymore).	“一会儿” (yí huìr) has no clear time duration. In different contexts, “一会儿” might refer to a few minutes, several minutes, or even longer. For instance, “我一会儿就到” (I'll be there in a moment), but the exact duration depends on factors like the speaker's distance or transportation mode. Until these are clarified, the meaning remains in a superposition state.
Arabic Family	قلب (qalb) can mean "heart," "center," or "core." In the sentence “هو في قلب المدينة” (He is in the heart of the city), without further context, قلب could refer to either "the center of the city" or	The sentence “رأيتُه قبل أن يذهب” (I saw him before he left) can be interpreted as "I saw him before he left" or "I saw him before he was about to leave." Without sufficient context, the sentence	The word كثير (kathir) means "a lot," but how much constitutes "a lot" is unclear. Depending on the context, it could refer to a different range of quantities (e.g., when referring to people vs. objects). Until context is

	"the heart (symbolically the most important part) of the city." Only with context can the exact meaning be determined.	remains ambiguous and needs additional temporal or event details to make the meaning clear.	provided, its meaning is in a superposition state.
Indo-European Family (German)	Strecke can mean "distance," "line segment," or "carriage (train car)." In sentences like "Wir haben eine lange Strecke gefahren" (We traveled a long distance) and "Dieser Wagen hat drei Strecken" (This car has three carriages), Strecke takes on different meanings. Without specific context, its meaning is in a superposition state and can only be determined by context.	The German sentence "Der Wagen muss repariert werden, der den Wagen traf" (The car that hit the car must be repaired) contains ambiguity in its structure due to the relative clause "der den Wagen traf" (that hit the car), leading to confusion about which car hit which. Without further clarification, the sentence remains in a superposition state, and meaning is determined by specific contextual information.	In German, viel means "many" or "a lot," but the specific quantity it refers to varies depending on context. For example, "Es gibt viel Schnee" (There is a lot of snow) vs. "Es gibt viel Geld" (There is a lot of money), where the standards of "much" differ depending on the context. Without clear quantification, its meaning remains in a superpositions.

4. The Phenomenon of Entanglement in Language: An Analogy Between Linguistics and Quantum Mechanics

At first glance, linguistics and quantum physics may seem entirely unrelated, yet through some profound analogies, striking similarities between the two emerge. Quantum entanglement is a unique phenomenon in quantum physics, referring to the deep correlation between two particles, such that their states instantaneously influence each other, no matter the distance between them (Yin et al., 2017). In linguistics, a similar phenomenon can be found: the interconnectedness and mutual influence of the meanings of words within a given context, creating a unique "linguistic entanglement." In everyday communication, language phenomena such as homophones, puns, jokes, and idiomatic expressions often rely on the tight connection between multiple meanings. In the case of puns, a single word can evoke multiple meanings simultaneously, while

the punchline of a joke "measures" or finalizes these ambiguities into a single interpretation. For example, the phrase "Time flies like an arrow" can be understood both as describing the speed of time passing and as a pun, drawing a comparison between time and an arrow in a more metaphorical sense. One part of the sentence influences the potential meanings of the other part, much like the interdependence between particles in a quantum system. This phenomenon of entanglement in language reflects the complex interactions between words and phrases, where meanings cannot exist in isolation but are deeply embedded in their context. In Chinese, for example, "树叶" (shù yè) can refer to a specific leaf or symbolize some aspect of life depending on the context, a type of polysemy similar to the phenomenon of puns in English, illustrating how semantics intertwine and entangle across different languages.

This phenomenon is reflected in various languages. For instance, in Chinese, "银行" can mean a financial institution or the bank of a river, depending on the context; in German, the sentence structure can cause verbs to appear at the end, such as in "Er hat das Buch gelesen" ("He read the book"), leading listeners to wait for the full context to understand the sentence's meaning; in Japanese, the use of particles makes the relationships between sentence components tighter, as seen in "私は本を読みます" ("I read books"), where the meaning of the sentence depends on the subtle change in the particle "を." In Russian, changes in verb prefixes can alter the entire meaning of a verb, such as "писать" ("to write") and "записать" ("to record"), which can only be fully understood in context; in French, due to the presence of multiple verb conjugations, the meaning of a verb often depends on tense and mood, as in "Il a conduit" ("He drove"), where the verb "conduire" takes on different meanings with changes in tense. Similarly, in Arabic, the meaning of a word can change drastically depending on the prefixes or suffixes attached to the root, such as "كتب" ("to write") and "مكتوب" ("written"), and, like in German, where the meaning of the verb is only clarified at the sentence's end, Arabic word meanings are fully understood only within their contextual framework. These examples illustrate how languages from different linguistic families

achieve meaning "entanglement" through context, much like the correlation of particle states in quantum entanglement.

Coherence in quantum mechanics refers to the tight connection between the states of multiple particles, a phenomenon that can also be found in language, in the way context influences word meanings. The meaning of a word often depends on the sentence structure and surrounding vocabulary. Take the English word "lead" as an example; depending on the context, it can refer to the metal "lead" or the action of "leading." This dependence on context for meaning changes mirrors the mutual dependence of particle states in quantum entanglement. Similarly, in Korean, the meaning of a sentence may hinge on the verb ending, such as in "나는 학교에 간다" ("I go to school"), where the verb ending "간다" carries the speaker's intent and the level of politeness, much like the use of modal verbs in English. In this case, context functions similarly to a quantum measurement, determining the final meaning of the word. Through these analogies, we gain a deeper understanding of language's polysemy, structure, and the pivotal role context plays in comprehension. This interdisciplinary study offers a fresh perspective on the potential connections between linguistics and physics, allowing us to understand human cognition's complexity and systematics from a deeper level. Linguistics research, beyond the study of language itself, provides crucial insights into how humans communicate and comprehend in a complex world. This offers a stimulating direction for future interdisciplinary exploration, enhancing our understanding of the interaction between language and the natural world.

5. Application of Quantum Models in Cognition and Language Processing

In the field of cognitive science, an increasing number of researchers are adopting quantum models to describe how the brain processes complex language and cognitive phenomena (Hameroff, 2012a, b). Traditional logic and classical probability theory often fall short when explaining how humans deal with ambiguity and uncertainty in language. This is because the vagueness and diversity within language cannot always

be captured by precise, deterministic probabilities. Quantum probability theory, however, has unique properties such as superposition and entanglement, which provide a more suitable framework (Primas, 1990) for explaining cognitive processes in the face of uncertainty, especially in decision-making and semantic processing. Quantum models have demonstrated significant advantages in experimental data related to these processes. Through quantum models, scientists can better simulate how the brain simultaneously evaluates multiple meanings of polysemous words, or how it dynamically updates meanings when confronted with complex linguistic structures—tasks that are difficult to achieve with traditional methods.

For example, in English, the word "bank" can simultaneously refer to a riverbank or a financial institution. When people hear this word, the brain holds both possible meanings until further contextual clarification is provided. For instance, "He is depositing money at the bank" clearly refers to the financial institution, while "He is walking along the riverbank" clearly refers to the riverbank. Similarly, in Chinese, the word "打架" (dǎ jià) can have different interpretations depending on the context. For example, in "打架是错误的" (Fighting is wrong), it refers to a physical action, whereas in "他们在打架与沟通之间犹豫" (They are hesitating between fighting and communicating), it symbolizes a state of conflict. These multiple meanings are analogous to quantum superposition states, where the meanings remain "superposed" until additional information collapses them into a single interpretation.

Furthermore, in German, the word "Schloss" can refer to either a "castle" or a "lock," a polysemy that requires the brain to retain multiple possibilities until more contextual information clarifies the specific meaning. In Spanish, "banco" can refer to a "bench" or a "bank," with the precise meaning depending on the context. These varieties of meaning also align with quantum models that describe superposition states.

Quantum decision theory further reveals how humans process ambiguous and conflicting information when making decisions (Yukalov, 2020). Unlike classical decision theory, people often maintain multiple conflicting ideas without a clear

conclusion—this state is similar to the superposition state in quantum physics. It is only when external factors or internal needs force a decision that these possibilities "collapse" into a specific outcome. This theory also applies to language processing: when people first hear or read a sentence, they may retain multiple possible interpretations until further contextual information resolves these ambiguities, ultimately selecting the single intended meaning. This quantum-like processing mechanism better reflects human dynamic adaptability to complex contexts and ambiguous information, illustrating the non-linearity and complexity of cognitive processes.

For instance, in French, the word "cour" can mean "court," "yard," or "course." The brain holds these possibilities in a superposition state until enough context clarifies the meaning. For example, "il joue dans la cour" can mean "He plays in the yard," while "il va à la cour" can mean "He goes to court." Similarly, in Japanese, the word "はし" (hashi) can mean "bridge" or "chopsticks," with the specific meaning depending on the sentence context. This uncertainty can also be viewed in quantum models as a superposition of different states, which only resolves through further information. These linguistic phenomena demonstrate how quantum models provide a more effective framework for describing how humans process polysemy, offering a more flexible and accurate reflection of real cognitive processes than traditional logic or classical probability theory.

6. Conclusion and Outlook

This paper explores the similarities between quantum mechanics and linguistics, revealing the potential connections between the two in handling uncertainty and complexity. Concepts such as discreteness, superposition, and quantum entanglement in quantum mechanics find analogies in phenomena like polysemy, context dependence, and lexical association in linguistics. This analogy provides a new theoretical framework for understanding the multi-layered structure and complexity of language, while also guiding the direction for future interdisciplinary research.

The discreteness in quantum mechanics, where microphysical quantities can only take specific discrete values, parallels the discrete structure of language. In linguistics, vocabulary and grammar rules are composed of discrete minimal units, such as phonemes, letters, and grammatical structures, allowing language to express an infinite range of meanings through a finite set of rules. The structure of English sentences demonstrates clear grammatical boundaries through the tense variations of verbs, while Chinese conveys complex meanings through word order and contextual relationships. Discreteness not only enables different language systems to meet diverse expressive needs but also maintains the logic and structure of language.

Moreover, the concept of quantum superposition offers an analogy for explaining ambiguity and polysemy in language. In quantum mechanics, a system can exist in multiple mutually exclusive states simultaneously and only "collapses" into a specific state upon measurement. Similarly, ambiguous sentences and polysemous words in language may retain multiple interpretations when the context is unclear. The specific meaning only emerges when "measured" by the context, which mirrors the collapse phenomenon in quantum systems. Quantum entanglement also finds an analogy in linguistics, such as the context dependence and interrelationship of words. Certain words or expressions can only be fully interpreted in specific contexts, much like the interdependent states in quantum entanglement.

The quantum-language analogy is not just a philosophical reflection; it also provides a novel framework for understanding the complexity and dynamics of language. By applying quantum mechanical principles to linguistics, scholars can simulate the multiple possibilities involved in language processing. Quantum decision theory is being used by cognitive scientists to explain how the brain maintains multiple interpretations when confronted with polysemy and complex sentences, until a clear meaning emerges. In the future, the intersection of quantum mechanics and linguistics could offer new insights into the quantum characteristics of human cognitive functions. By exploring the potential reflection of quantum effects in language processing,

particularly in the handling of ambiguity, polysemy, and context changes, the interdisciplinary application of quantum mechanics will offer fresh perspectives in linguistics, cognitive science, and social science research.

Such interdisciplinary research not only deepens our understanding of the complexity of language systems but also uncovers the underlying mechanisms of human thought and cognition, opening new pathways for the exploration of the mysteries of human thought and language.

References

Ball, P., 2011. Physics of life: The dawn of quantum biology. *Nature* 474, 272-274.

Bowern, C., 2015. Linguistics: evolution and language change. *Curr Biol* 25, R41-43.

Cai, J., 2016. Quantum biology: explore quantum dynamics in biological systems. *Science China Information Sciences* 59.

Cao, J., Cogdell, R.J., Coker, D.F., Duan, H.G., Hauer, J., Kleinekathofer, U., Jansen, T.L.C., Mancal, T., Miller, R.J.D., Ogilvie, J.P., Prokhorenko, V.I., Renger, T., Tan, H.S., Tempelaar, R., Thorwart, M., Thyryhaug, E., Westenhoff, S., Zigmantas, D., 2020. Quantum biology revisited. *Sci Adv* 6, eaaz4888.

Diamond, J., 2011. Linguistics: Deep relationships between languages. *Nature* 476, 291-292.

Fitch, W.T., 2007. Linguistics: an invisible hand. *Nature* 449, 665-667.

Galofaro, F., Doan, B.-L., Toffano, Z., 2016. Linguistics and Quantum Theory: Epistemological Perspectives, 2016 IEEE Intl Conference on Computational Science and Engineering (CSE) and IEEE Intl Conference on Embedded and Ubiquitous Computing (EUC) and 15th Intl Symposium on Distributed Computing and Applications for Business Engineering (DCABES), pp. 660-668.

Greenberg, J.H., 1959. Current trends in linguistics. *Science* 130, 1165-1170.

Hameroff, S., 2012a. How quantum brain biology can rescue conscious free will. *Front Integr Neurosci* 6, 93.

Hameroff, S., 2012b. Quantum brain biology complements neuronal assembly approaches to consciousness Comment on "Consciousness, biology and quantum hypotheses" by Baars and Edelman. *Physics of Life Reviews* 9, 303-305.

Hruschka, D.J., Branford, S., Smith, E.D., Wilkins, J., Meade, A., Pagel, M., Bhattacharya, T., 2015. Detecting Regular Sound Changes in Linguistics as Events of Concerted Evolution. *Current Biology* 25, 1-9.

Kim, Y., Bertagna, F., D'Souza, E.M., Heyes, D.J., Johannissen, L.O., Nery, E.T., Pantelias, A., Sanchez-Pedreño Jimenez, A., Slocombe, L., Spencer, M.G., Al-Khalili, J., Engel, G.S., Hay, S., Hingley-Wilson, S.M., Jeevaratnam, K., Jones, A.R., Kattnig, D.R., Lewis, R., Sacchi, M., Scrutton, N.S., Silva, S.R.P., McFadden, J., 2021. Quantum Biology: An Update and Perspective. *Quantum Reports* 3, 80-126.

Koch, H.J., 2017. Quantum Biology: Unit Membrane Reduces Entropy Due To Wave Particle Duality. *Neuroquantology* 15, 42-44.

Levy, A., 2021. How quantum biology could help birds 'see' magnetic fields. *Nature*.

Primas, H., 1990. Mathematical and Philosophical Questions in the Theory of Open and Macroscopic Quantum Systems, in: Miller, A.I. (Ed.), *Sixty-Two Years of Uncertainty: Historical, Philosophical, and Physical Inquiries into the Foundations of Quantum Mechanics*. Springer US, Boston, MA, pp. 233-257.

Schrodinger, E., 1944. What is life?

Trixler, F., 2013. Quantum Tunnelling to the Origin and Evolution of Life. *Current Organic Chemistry* 17, 1758-1770.

Vattay, G., Kauffman, S., Niiranen, S., 2014. Quantum Biology on the Edge of Quantum Chaos. *Plos One* 9.

Wuethrich, B., 2000. Linguistics. Learning the world's languages--before they vanish. Science 288, 1156-1159.

Yin, J., Cao, Y., Li, Y.H., Liao, S.K., Zhang, L., Ren, J.G., Cai, W.Q., Liu, W.Y., Li, B., Dai, H., Li, G.B., Lu, Q.M., Gong, Y.H., Xu, Y., Li, S.L., Li, F.Z., Yin, Y.Y., Jiang, Z.Q., Li, M., Jia, J.J., Ren, G., He, D., Zhou, Y.L., Zhang, X.X., Wang, N., Chang, X., Zhu, Z.C., Liu, N.L., Chen, Y.A., Lu, C.Y., Shu, R., Peng, C.Z., Wang, J.Y., Pan, J.W., 2017. Satellite-based entanglement distribution over 1200 kilometers. Science 356, 1140-1144.

Yukalov, V.I., 2020. Evolutionary Processes in Quantum Decision Theory. Entropy (Basel) 22.